

PALASH PANCHAL

AI & Autonomous Systems Engineer

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AI engineer who ships multi-agent systems into production — model routing and context economics, MCP connectors and OAuth 2.1, containerized agent execution, retrieval, and the eval gates that keep it honest. Built the security-compliant enterprise AI platform used by **5,000 employees**: **+70% work efficiency, 40% less delivery dependency and duration**.

EXPERIENCE

AI Engineer — Enterprise AI Platforms *Nouryon Chemicals Jul 2024 – Present*

- **Curiosity** — built the Azure-hosted AI workspace that replaced commercial LLM tooling the company could not licence under its data-residency rules. **5,000 users; +70% work efficiency and 40% less delivery dependency and duration**, by collapsing manual hand-offs between engineers, support desks, and enterprise systems into single prompted workflows.
- Engineered the **model router** and context layer over open-source models — per-request model tiering, conversation compaction, cache-stable system prefixes — holding long sessions inside context limits without losing task state, at zero per-seat licence cost.
- Shipped **MCP-protocol connectors** for **ServiceNow** and **SAP** under **OAuth 2.1 with dynamic client registration**, so agents act as the signed-in user under delegated scopes, not a shared service principal.
- Built a **Deep Research MCP server** running **7 agents asynchronously**, and **Capabilities**, a runtime executing .md/.skill definitions (Claude-Skills compatible) in **Azure Container Apps session pools** from **ACR** images, generating templated corporate documents through those connectors.
- Delivered **Helix**, a browser VS Code IDE on the same router and containers so users choose their execution environment on the web; **Projects** workspaces over **Azure AI Search** agentic retrieval; and autonomous agents completing Microsoft 365 tasks under per-user **Graph** tokens.
- **Symphony** — designed a prompt-to-application pipeline, invoked from Curiosity, letting **non-technical business users** self-serve small applications from one prompt: it decomposes the request, formalizes intent, runs business- and problem-statement impact analysis, drafts a **draw.io** Azure architecture, then hands the package to containerized Claude CLI agents backed by a component library of build/test/verify agents, domain and model-orchestrator skills, and **Bicep** modules. **+30% process efficiency** on shipped automations.

Test Analyst Intern *Nouryon Chemicals Feb 2024 – Jun 2024*

- Engineered a data-quality pipeline ingesting **SAP, Cosmos DB, and OneHub** into the Azure Data Platform, validating transformed output against source through Blob Storage and **Azure DevOps YAML**, and paging the owning developer on mismatch via **Microsoft Graph**.
- Automated monthly **Power BI** PBIX reporting and production API test suites (Python, Postman) instrumented with **Application Insights**, eliminating a recurring manual reporting cycle.

Machine Learning Intern, Verzeo · Jul–Sep 2022 · customer-churn exploratory analysis and predictive modelling in a small team.

SELECTED PERSONAL PROJECTS

SARA — Multi-Agent Wellbeing Companion *Python, FastAPI, SSE, SQLite, Fernet, Kokoro ONNX · non-profit · PWA + app stores planned*

- Built a **4-agent** pipeline to make first-line support reachable when a friend or therapist isn't — explicitly a bridge to human help, never a replacement: safety triage on every message (4-level risk enum via JSON-schema structured output), a silent formulation analyst, a streaming companion, rolling memory — over **24 API endpoints, 11 field-encrypted SQLite tables, and 32 knowledge areas**.
- Cut perceived latency with **speculative execution**: a debounced background thread pre-runs triage and generation against the user's in-progress draft so the reply streams the instant they send. **High-risk turns bypass the model entirely** to a deterministic helpline handoff.
- Engineered the token economics and release gate — prompt caching on a byte-stable prefix, rolling summarization every 10 turns, per-agent cache-hit accounting — plus **90 red-team crisis cases, 14 jailbreak cases, and 11 LLM-judged personas that fail the build** if triage fails open or misses a high-risk message.

SECOND BRAIN — Local-First Developer Knowledge Graph *Python, SQLite/FTS5, ONNX embeddings, MCP*

- Built a knowledge layer that stops agents burning tokens re-scanning repos: **7 collectors** (git across **12 repos**, Claude Code sessions, VS Code, shell, browser, markdown) feed an Obsidian vault and auditable SQLite graph, on-device, with **9 secret-redaction patterns** at the ingest boundary.
- Designed a **6-signal rules matrix** scoring every item into 4 retention tiers, **filtering ~80% of activity before any model call** (measured over 639 logged decisions), each keep/drop stored with its score and reason for audit.
- Implemented hybrid retrieval — **FTS5 BM25 fused with on-device 384-dim BGE embeddings** for zero-token semantic search — behind **6 MCP tools** that return ~200-char pointers instead of documents, under hard **8K-char input and 2-call/day** budgets with SHA-1 response caching.
- **In progress**: user-curated add/remove of graph regions, and shareable per-repo context packs so a second developer inherits the graph instead of re-deriving it.

TECHNICAL SKILLS

AI & Agents: multi-agent orchestration, MCP (client, server, connectors), OAuth 2.1 + dynamic client registration, model routing, context compaction, prompt caching, hybrid retrieval (BM25 + vector), agentic retrieval, structured output, LLM evaluation and red-teaming

Azure: AI Foundry, Container Apps (session pools), Container Registry, AI Search, Document Intelligence, Cognitive Services, Key Vault, Cosmos DB, Blob Storage, Data Platform, DevOps YAML, Application Insights, Load Balancer, Bicep, service principals and managed identity, Microsoft Graph API

Engineering: Python, FastAPI, JavaScript, Node.js, Express.js, Java, SQL, SQLite/FTS5, MongoDB, REST, SSE, asyncio and threading, containers, Postman, Power BI, Tableau

EDUCATION

K. J. Somaiya College of Engineering, Mumbai — B.Tech Computer Engineering, Honours in Data Science · 2024 · CGPA 8.078